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18 November 1975
SUPERSEDING
MIL-N-36124A (DM)
25 February 1965

MILITARY SPECIFICATION

NEEDLES, HYPODERMIC, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, disposable, hypodermic needles.

1.2 Classification. Needles shall be of the following sizes, (gages and lengths), as specified (see 6.1):

Gage	Length
15	1-1/2
18	1-1/2
19	1-3/4
20	1-1/2
21	1-1/4
22	1
23	3/4
23	1
24	1-1/2
25	1/4
25	5/8
26	1/2
26	1-5/8
27	1/2

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Federal

MMM-A-250	- Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-B-676	- Box, Set-up.
PPP-T-76	- Tape, Pressure-sensitive Adhesive Paper, Water-Resistant, (For Carton Sealing).

Military

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-S-36586	- Sterlization Strip Test, Bacterial Spore.

STANDARDS

Federal

FED-STD-151	- Metals; Test Method.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

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American Iron and Steel Institute
Steel Products Manual, Stainless and Heat-Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI Z70.1 — American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

U. S. Pharmacopeial Convention, Inc.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Cannula. The cannula shall be fabricated from either seamless drawn; or welded and drawn, stainless steel tubing, American Iron and Steel Institute (AISI) type 304 stainless steel. Material shall be tested, as specified in 4.4.10, for chemical composition.

3.1.2 Hub. The hub shall be made from a suitable plastic compound containing no elutable toxic materials, of suitable aluminum alloy, nickel-plated brass, or nickel-chromium plated brass. The hub shall be compatible with known medications. Aluminum hubs shall be degreased and anodized or chemically passivated, resulting in a nontinted finish. Also needles with aluminum hubs shall withstand an exposure to sodium chloride as required in 3.2.2.1.

3.1.3 Metal insert. If a metal insert is used (to connect the hub to the cannula), it shall be fabricated from nickel plated brass, nickel plated copper alloy, suitable aluminum alloy (degreased and chemically passivated or anodized), or from stainless steel.

3.1.4 Lubricant. The lubricant on the exterior of the cannula shall be inert, non-toxic silicone compound suitable for the specific use. The silicone compound shall be pharmaceutical grade.

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3.2 Style, design, and construction. Shall be a sterile, hypodermic needle, with a regular bevel point, packaged in an individual, sealed, sterile package as hereinafter specified. This sealed package shall maintain the sterility of its contents, and shall be so designed as to permit ready dispensing of the needle by sterile techniques.

3.2.1 Cannula. The cannula shall be a straight tube with a regular bevel point on the distal end. The point shall be ground concave or flat and may have a center ridge at the distal end to strengthen the tip. The point shall be so ground at the outer end as to enhance the keeness of the tip and cutting edges. Both outer and inner edges of point shall be free from burrs and slivers. The heel of the point shall be dull to prevent coring. The tip of the point shall be straight and strong, and of maximum keeness. The proximal end of the cannula shall extend well into the hub sufficiently to allow for secure staking. The cannula shall be centered in the distal end of the hub in such a manner as to provide a straight continuous fluid path from the hub cavity. The cannula shall not deviate from the central longitudinal axis of the needle by more than 3 degrees. The exterior of the cannula shall be clean, smooth and free from pits, deep furrows and die marks.

3.2.1.1 Dimensions (cannula and point). Dimensions of cannula (other than length) and point shall be as indicated in table I for the appropriate gage. The cannula length shall be as specified (see 1.2 and 6.1). The length shall be measured from the tip of the point to the junction with hub. Tolerances for cannula length shall be indicated in table II.

TABLE I. Dimensions of cannula and point

Gage	Cannula		Point
	Outside diameter, limits (inch)	Inside diameter, minimum (inch) 1/	Length, limits (inch) 2/
27	0.0155-0.0165	0.0073	0.053-0.086
26	0.0175-0.0185	0.0093	0.059-0.095
25	0.0195-0.0205	0.0093	0.078-0.103
24	0.0215-0.0225	0.0113	0.092-0.112
23	0.0245-0.0255	0.0125	0.105-0.129
22	0.0275-0.0285	0.0155	0.120-0.146
21	0.0315-0.0325	0.0195	0.133-0.163
20	0.0340-0.0360	0.0220	0.142-0.180
19	0.0405-0.0435	0.0255	0.175-0.215
18	0.0475-0.0505	0.0315	0.203-0.249
15	0.070 -0.074	0.048	0.288-0.352

1/ Measured at point.
2/ Measured from heel to tip parallel with longitudinal axis of cannula.

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TABLE II. Tolerances for cannula length

Length (inch)	Tolerance (inch)
Up to 1/2	+ 1/32, -1/16
From over 1/2 up to 3/4	+ 3/64, -1/16
From over 3/4 up to 1-1/2	± 1/16

3.2.1.2 Corrosion resistance of cannula. When tested as specified in 4.4.9, cannula shall show no appreciable corrosion.

3.2.2 Hub. Shall be Luer-lock hub to provide means for facilitating attachment to, and removal from, a syringe so designed and constructed to form a leak-proof connection with male Luer lock connection constructed in accordance with Figure 4 of American National Standards Institute (ANSI) Z70.1, and shall make a mechanically stable, fluid tight connection (see 3.6) with a syringe having a Luer lock tip. The tapered portion of the hub cavity shall be 0.060 inch per inch in accordance with Figure 1 of ANSI Z70.1. The hubs shall be tested for diameter and depth of hub cavity as specified in 4.4.1. Hubs which do not fit the plug gage within the "GO" - "NO GO" limits specified in 4.4.1 shall be considered not acceptable. The fit of the Luer taper of plastic hubs on the plug gage need not be tested with the feeler gage present. When the hub is fitted onto the plug gage as specified in 4.4.1, there shall be no movement of the hub on the gage when the hub is pressed lightly from side to side in all directions. If hub is made of brass, the entire hub cavity wall, down to the entrance of the lumen, as well as the exterior of the hub shall be well plated with nickel or chromium over nickel. Similarly, if fabricated of aluminum, the anodization or chemical passivation shall include the entire hub cavity wall, down to the entrance of the lumen, as well as the exterior of the hub. A metal insert may be used for connecting the hub to the cannula. If the insert is fabricated from brass or copper alloy (see 3.1.3), the entire insert cavity wall, down to the entrance of the cannula, as well as the exterior of the insert, shall be well plated with nickel or chromium over nickel. Hub and cannula shall be securely assembled. Soft solder (lead-tin alloy) shall not be used in joining the cannula and hub.

3.2.2.1 Corrosion resistance of aluminum hubs. Needles with aluminum hubs shall have the corrosion resistance of the aluminum hubs tested by a one-hour exposure to 3% sodium chloride without visible change as specified in 4.4.12.

3.3 Freedom from foreign matter. When tested as specified in 4.4.2, the hub cavity shall be free from chips and other foreign matter. In addition, a visual examination of the lumen of the cannula shall reveal no presence of foreign matter, chips, burrs, or occlusion of the lumen.

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3.4 Security of staking. When tested as specified in 4.4.3, the cannula shall not loosen or pull out of the hub.

3.5 Resistance to breakage. When tested as specified in 4.4.4, there shall be no breakage.

3.6 Fluid tightness. When tested as specified in 4.4.5, the connection of the cannula with the hub, and the connection of the hub with the test gage, shall not show leakage exceeding the formation of a falling drop.

3.7 Sterility. The needles shall be sterile when tested with a biological indicator as specified in 4.4.6.

3.7.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.7.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and subsequently sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 3.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

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3.8 Freedom from pyrogens. When tested as specified in 4.4.7, needles shall comply with the pyrogen test requirements of the USP.

3.9 Safety. When tested as specified in 4.4.8, there shall be no deaths or discernible symptoms of toxicity of the test animals in determining acute toxicity or adverse tissue reaction. The injection sites on the test animal shall be free from evidence of irritation or toxicity (trauma shall be ruled out with suitable saline control sites).

3.10 Stiffness. When tested as specified in 4.4.11, the maximum deflection of the cannula shall not exceed the distance indicated in table III for appropriate gage.

TABLE III. Stiffness test requirements

Gage	Maximum deflection (inch)
27	0.022
26	0.019
25	0.026
24	0.041
23	0.033
22	0.029
21	0.016
20	0.021
19	0.033
18	0.017
15	0.006

3.11 Certificates. For destination, inspection only, certificates of freedom from pyrogens, and safety, as specified below, shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection, is made at source certificates of freedom from pyrogens and safety shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.11.1 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens, as specified in 3.8 and 4.4.7, and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.11.2 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.9 and 4.4.8, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

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3.12 Workmanship. Workmanship shall be first class throughout. Needles shall be free from defects which detract from their appearance or may impair their serviceability.

3.13 Precautionary information (plastic hub needles only). For needles with plastic hubs, each unit package shall be provided with information cautioning the user against the use of parenterals, solution, etc. that would be harmful to the plastic.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic and metals may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105 and, unless otherwise specified, shall be based on the unit of product being one needle (end item).

4.2.2 Sampling for examination. A sample of needles shall be selected from each lot in accordance with MIL-STD-105. The inspection level and Acceptance Quality Level (AQL) shall be as indicated in table IV.

TABLE IV. Sampling for examination

Type defect	Inspection level 1/	AQL (percent defective)
Major	I	1.0
Minor	I	4.0

1/ Use double sampling plan for normal inspection.

4.2.3 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table V.

TABLE V. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Dimensions	1.2 and 3.2.11	---	S-1	1.5
Composition of cannula	3.1.1	4.4.10	S-1	1.5
Corrosion resistance of cannula	3.2.1.2	4.4.9	S-2	1.0
Corrosion resistance of aluminum hubs	3.2.2.1	4.4.12	S-2	1.0
Freedom from foreign matter	3.3	4.4.2	S-3	1.5
Security of staking	3.4	4.4.3	S-1	1.5
Resistance to breakage	3.5	4.4.4	S-1	1.5
Fluid tightness	3.6	4.4.5	S-3	1.5
Sterility	3.7	4.4.6	S-3	1/
Freedom from pyrogens	3.8	4.4.7	S-1	1/
Safety	3.9	4.4.8	S-1	1/
Stiffness	3.10	4.4.11	S-1	1.5
Gaging	3.2.2	4.4.1	S-1	1.0

1/ Acceptance number shall be zero for all sample sizes.

4.2.4 Where possible, the same sample unit shall be used for the determination of two or more test characteristics

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4.3 Examination. The needles shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables IV and VI. Examination, however, shall not be restricted to defects listed in table VI. Any needles in the sample containing one or more defects shall be rejected and, if the number of defective needles in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE VI. Classification of defects

Categories	Defects
Critical:	None defined
Major	
101	Cannula lumen not free of obstructions or foreign matter.
102	Cannula lumen not free of burrs and slivers.
103	Cannula outer surface not smoothly finished or free of pits, tool marks, burrs, scale, or corrosion.
104	Cannula not free of cracks.
105	Cannula not straight.
106	Cannula exterior not free from foreign matter.
107	Hub not free of cracks or flash.
108	Hub not free of burrs, slivers, or foreign matter.
109	Point missing.
110	Point blunt or bent, 1/ not sharp, not symmetrical or properly formed, or not free of burrs, feathered edges, or other detrimental defects (examined under 7X magnification).
Minor	
201	Discoloration of outside surface of hub.

1/ A bent point is defined as any deviation of the point from the longitudinal axis of the needle when viewed at 7X from the side of the needle and in a line parallel to the plane of the bevel.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Diameter and depth of hub cavity. Hub cavity shall be tested on the male Luer taper gage in accordance with plug gage of Figure 5 of ANSI Z70.1. (This gage shall be as shown in ANSI Standard, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.5). The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with Figures 6 and 8 of ANSI Z70.1. The fit of the Luer taper of plastic hubs on the plug gage need not be tested with the feeler gage present. All hubs shall provide a firm, non-wobble fit when testing the Luer taper of hubs on the plug gage. The flange on the Luer hub shall be properly recessed from the proximal edge and of dimensions to provide a full, firm, leak-proof fit of the hub and tip (see fig. 3 of the ANSI Standard Z70.1).

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4.4.2 Freedom from foreign matter test. The hub cavity shall be examined under 7X magnification, and good illumination, for chips and other foreign matter. In addition, a cotton swab tightly wrapped on the end of a small wooden applicator shall be turned down into the bottom of cavity, withdrawn, and examined for chips and other foreign matter under 7X magnification and good illumination. The cannula shall be tested for blockage by inserting a wire through the lumen of the cannula. In addition, the needle shall be fitted to a 10cc syringe. The syringe shall be filled with distilled water by pouring the water into the barrel. Insert the piston and eject the distilled water into a medium texture, quantitative filter paper properly inserted into a 60°, 75 mm, smooth glass funnel. As a blank, take an equal amount of distilled water and pour it into a similar, separate funnel filter paper arrangement as used for the syringe. Examine the residue on the filter papers at 7-power magnification, after the water has passed through them, and determine if foreign material ascribable to the needle is present.

4.4.3 Security of staking. A straight pull of 5-pounds shall be applied between the cannula and hub for a period of 1-minute.

4.4.4 Resistance to breakage test. The needle hub shall be held in a fixed position, and a force shall be applied to the cannula, at a distance from the hub as indicated in table VII, in order to bend the cannula in a clockwise direction through an arc of 25 degrees from the original or neutral position. The operation shall then be reversed (i.e., bent through an arc of 25 degrees from the neutral position in a counter-clockwise direction), and finally, the cannula shall be returned to the neutral position. This complete operation (or cycle) shall be repeated 20 times.

TABLE VII. Breakage test 1/

Gage	Contact distance from hub (inch)
27	5/16
26	5/16
25	7/16
24	1/2
23	5/8
22	11/16
21	3/4
20	1
19	1-1/8
18	1-1/4

1/ The 15-gage needle shall not be subjected to this test.

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4.4.5 Fluid tightness test. With the needle in a horizontal position, the hub shall be attached by means of a plug gage as shown in Figure 5 of ANSI Z70.1 (except that gage shall have been provided with a fluid path) to a water pressure source having a pressure gage. The needle shall be secured to the Luer plug gage by applying it with a 3-1/2 pound axial force and while maintaining this compressive stress twisting the needle hub 45° relative to the plug gage. The water shall be permitted to run through the needle so as to fill any hub cavity space and the lumen of the cannula. The lumen shall then be sealed by a piece of solid rubber pressed on to the point. A pressure of 45 pounds per square inch gage (p.s.i.g.) shall be applied for 10 seconds.

4.4.6 Sterility test. The needles shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The inucubation temperature shall be as indicated in paragraph 4.4.6.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in table V, Sampling Criteria for Tests.

4.4.6.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus	55° - 65°C
	B. subtilis var. niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.6.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.6 Sterility Test and 4.4.6.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.6.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

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4.4.6.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.6.4 Product testing. Testing for sterility of the product after receipt by the Government shall be without a biological indicator and in accordance with the USP.

4.4.7 Freedom from pyrogens test. Place 10 needles in a sterile, pyrogen-free flask, pour 250 milliliters (ml.) of sterile, pyrogen-free saline test solution (USP) into the flask and then heat the contents of the flask at not less than 40° Centigrade (C.) nor higher than 50° C. for 1 hour. The resulting saline extract shall be submitted to the "Pyrogen Test" as described in the USP.

4.4.8 Safety tests.

4.4.8.1 Needles with plastic hubs. Needles furnished with plastic hubs shall be tested for safety as follows:

Cover 20 needles with 30 ml. of sterile, pyrogen-free saline test solution (USP) and warm at 37° C. for 24 hours. Decant the saline and test, undiluted, for both acute toxicity and tissue sensitivity. Determine acute toxicity by injecting 1.0 ml. of the eluate intravenously into each of 10 nonfasted, healthy 28 - to 32 - gram mice. Observe the animals at 4, 24, and 72 hours after injection for any deaths or discernible symptoms of toxicity. Determine tissue reaction by injecting 0.20 ml. of the eluate into each of 10 sites on the back of a rabbit: the injections shall be made in such a manner as to raise a distinct bleb. Observe the injection sites at 4, 24, and 48 hours after injection for any evidence of redness, swelling, or hardness. Trauma shall be ruled out with suitable saline control sites.

4.4.8.2 Needles with metal hubs. Needles furnished with metal hubs shall be tested as specified in 4.4.8.1, except that 10 needles shall be covered with 50 ml. of the saline test solution (specified in 4.4.8.1) and the solution containing the needles shall be boiled for 4 hours.

4.4.9 Corrosion-resistance of cannula. The cannula shall be immersed in 10 percent citric acid at room temperature for 5 hours and, after removal from the acid, shall be boiled in distilled water for 30 minutes and, then shall remain immersed for 48 hours in the distilled water. At the end of the 48-hour period, the cannula shall be allowed to dry by evaporation, and shall then be examined for evidence of corrosion. A glass vessel shall be used throughout this test.

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4.4.10 Chemical analysis. Cannula material (see 3.1.1) shall be subject to chemical analysis in accordance with method 111 or 112 of FED-STD-151.

4.4.11 Stiffness test. The cannula shall be placed across two test edges (0.002 inch radius maximum) separated by the distance indicated in Table VIII for appropriate gage. A load, as indicated in Table VIII shall be applied at the center of the span and the resulting maximum deflection of the cannula shall be measured using a gage with graduations of at least 0.001 inch. The force shall be applied through an edge with a maximum radius of 0.036 inch, and centered to within ± 0.003 inch.

TABLE VIII. Stiffness test

Gage	Separation of test edges, inch (± 0.003 inch)	Load, ounces (± 1/4 ounce)
27	0.375	20
26	0.375	25
25	0.500	25
24	0.500	40
23	0.625	40
22	0.625	50
21	0.625	50
20	0.750	50
19	1.000	70
18	1.000	70
15	1.000	90

4.4.12 Corrosion resistance of aluminum hubs. Needles with aluminum hubs shall be placed in a test tube after the cannula has been snipped off to leave a 1/16 - inch distal protrusion from the hub. The open end of the hub shall be up. The hub shall then be covered with 3% sodium chloride solution at 20° to 25° C (68° to 77° F). The NaCl shall be ACS reagent grade and the water shall be distilled. Exposure shall be maintained for one hour. During this exposure there shall be no discoloration visible or any sign of reaction.

4.4.13 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IX. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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TABLE IX. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One box containing one hundred needles, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Individual package. Each needle shall be packaged in an individual container of one of the following designs. The individual container design with needle protector for type I package, or without needle protector for type II or type III package, shall protect the needle point from damage and permit ready opening. The package design shall be such as to permit ready dispensing of the needle from the package, by sterile techniques and by attaching the syringe to the hub of the needle without touching the needle with hands or other instruments. The completed sealed individual package shall be substantially free from foreign matter and shall maintain the degree of sterility as specified in 3.7 until such time as seal is broken or package is opened.

5.1.2.1.1 Type I package. Blister pack, nonconforming, shall be fabricated with a plastic coated or uncoated (inside) paper backing and a transparent or translucent plastic bubble firmly adhered to the paper backing. Each cannula shall be protected with a thermo-plastic sheath which shall extend approximately two thirds onto the needle hub.

5.1.2.1.2 Type II package. Blister pack, conforming, shall be fabricated with a transparent plastic or plastic coated (inside) paper backing with a transparent or translucent plastic bubble which shall conform to the contour of the needle and maintain the hub of the needle in proper position for dispensing.

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5.1.2.1.3 Type III package. Rigid plastic container shall be extruded or molded in two parts of a thermo-plastic resin. The lower part of the container shall be designed with interior fittings to maintain the needle in a vertical position with sufficient area of the hub exposed to permit the attaching of the syringe. The upper part or cap shall fit over the lower part in such a manner as to permit sealing of the two parts. Sealing of the two parts shall be accomplished by means of a thermo-plastic area approximately 1/4 inch wide, in such a manner that it shall be tamperproof (i.e., an opened container, broken seal, shall be easily separated from the lower part without exerting any effort, and shall be cause for rejection).

5.1.2.2 Unit package.

5.1.2.2.1 Type I and Type II packages. Individual packages shall be furnished in strips, not exceeding ten (10) packages per strip. The strips shall be designed to permit removal of individual package from the strip without damage to the remaining packages. A number of strips containing one hundred (100) needles shall be packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676, except that board thickness of PPP-B-566 box shall be not less than .016. Closure shall be adequate to prevent accidental opening.

5.1.2.2.2 Type III package. One hundred (100) individual packages shall be packaged in a one-piece paperboard box of appropriate size and design. The individual packages shall be oriented within the box in a neat manner. Closure shall be secured to prevent accidental opening.

5.1.2.3 Intermediate package. Ten unit packages containing needles of one gage and length only shall be in a snug fitting box constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Five intermediate packages containing needles of one gage and length shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.2 Level B. Five intermediate packages containing needles of one gage and length shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed in accordance with method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual package. In addition to any commercial markings, each package shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act. In addition, the individual package shall be marked with the name, and size of the commodity contained, the sterility lot (control) number, and the legends: "STERILE" and "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement. As an alternate, the sterility guarantee may be shown on the unit package. If the legend is placed on the unit package, it shall be shown as: "Sterility of inner individual packages guaranteed unless individual inner packages are opened or damaged".

5.3.2 Unit package and intermediate package. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Sterility lot (control) number, contract or purchase order number and name of contractor shall be shown. In addition, the method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-9/73-G". Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot (control) number and the method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number, and date of the specification.
(b) Gage and length required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable National Stock Number.

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6.2 National Stock Number (NSN Coverage). This specification includes the following items appearing in the Federal Supply Catalog:

NSN	Item Identification
6515-00-655-5751	NEEDLE, HYPODERMIC, Disposable, 25 Gage, 5/8 inch, 100s
6515-00-754-2834	NEEDLE, HYPODERMIC, Disposable, 18 Gage, 1-1/2 inch, 100s
6515-00-754-2835	NEEDLE, HYPODERMIC, Disposable, 22 Gage, 1 inch, 100s
6515-00-754-2836	NEEDLE, HYPODERMIC, Disposable, 20 Gage, 1-1/2 inch, 100s
6515-00-754-2837	NEEDLE, HYPODERMIC, Disposable, 26 Gage, 1/2 inch, 100s
6515-00-754-2838	NEEDLE, HYPODERMIC, Disposable, 21 Gage, 1-1/4 inch, 100s
6515-00-754-2839	NEEDLE, HYPODERMIC, Disposable, 23 Gage, 3/4 inch, 100s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing acitvity:
Defense Supply Agency - DM

Project No. 6515-1939
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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-N-36124B NEEDLES, HYPODERMIC, DISPOSABLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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